

Sherbrooke Sensor Dashboard

An interactive Streamlit dashboard for environmental sensor monitoring and anomaly detection.

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Overview

This project turns a stream of environmental sensor readings — temperature, humidity, moisture, and gas levels — into an explorable web dashboard. Built with Streamlit, it loads a synthetic Sherbrooke sensor dataset from Kaggle, cleans and timestamps the records, and presents interactive trend, correlation, and anomaly views. It pairs classical statistical detectors with a trained machine-learning model so anomalies can be surfaced and inspected in context.

Goals

Deliver a self-contained, interactive monitoring tool that makes multi-variable environmental sensor data easy to explore, and that flags anomalous readings reliably using both statistical and machine-learning methods.

Objectives

- Ingest and preprocess two sensor datasets (a clean baseline and an anomaly-injected variant), indexing by timestamp and handling missing or malformed records.
- Provide interactive views for gas-level trends, environmental variables, seasonal patterns, and cross-variable correlations over user-selectable timeframes.
- Offer three anomaly-detection methods — IQR, Z-Score, and a trained XGBoost model with a tuned decision threshold — applied across the full series or a custom date range.
- Expose statistical summaries and downloadable reports through a clean sidebar-driven interface.

Approach

The application is structured as a single Streamlit app. Data is cached after download and preprocessing for responsiveness. Visualisations are built with matplotlib, seaborn, and Altair, and the anomaly-detection module loads a serialised XGBoost model and its tuned threshold to score readings, overlaying detected anomalies directly on the time-series charts alongside the statistical detectors.

Outcomes

- A working, end-to-end dashboard that monitors four environmental signals and detects anomalies interactively.
- Three complementary anomaly-detection strategies the user can switch between, combining interpretable statistical rules with a learned model.
- A reusable, configuration-light tool that runs locally from a single command and sources its data directly from Kaggle.